

Source A — Findings Report

Wikipedia Intro-Text County Embeddings (BAAI/bge-m3, LLM-cleaned)

Macro Embeddings project — internal technical report

Prepared 2026-07-11 · Data as of 2026-07-10 (post LLM-cleaning adoption)

1. Executive Summary

Source A embeds the Wikipedia introductory section for all 3,144 US counties and county-equivalents (BAAI/bge-m3, 1024-dim, L2-normalized), after passing the raw text through a local LLM boilerplate-cleaning step (gemma2:9b) adopted on 2026-07-10. Coverage is complete — the original 56-county gap (Virginia independent cities plus 19 naming mismatches) has been fully backfilled.

The embeddings carry a real but small geographic signal, and no economically meaningful structure. Specifically:

- Geography vs. similarity: a weak but statistically significant negative correlation (Mantel $r = -0.236$, $p = 0.002$) — nearby counties are only slightly more textually similar than distant ones. LLM cleaning roughly doubled this effect versus the prior regex-cleaned baseline ($r = -0.106$).
- Principal component 1 explains just 3.9% of variance, but does carry a real, interpretable theme: presence of founding/namesake narrative (e.g. "named for General X") vs. bare demographic-stub content. This is heavily concentrated in Texas counties (21 of the top 30 PC1 counties are Texan, against an 8.1% base rate) and reflects a Wikipedia editorial convention, not an economic or geographic pattern.
- K-means clustering finds only a weak two-way split (silhouette ≤ 0.028) — highly reproducible across random seeds (ARI ≈ 0.98) and statistically real (permutation $p \leq 0.001$), but small in magnitude and should not be described as regional structure.
- Embedding more article text (additional sections, or an economy-only section) does not improve county differentiation — tested and closed as a dead end.
- Most importantly for planning: the original E_macro proposal justified Source A on the basis that intro text carries distinctive economic-transition narrative. Unsupervised analysis does not support that — the strongest structure found is a Wikipedia editorial artifact, not economic content. This does not mean Source A should be dropped (its weak geographic signal is real, and correlation against actual economic variables from Sources B/E is still untested), but it is not yet validated for the role the proposal describes.

Bottom line: Source A is technically sound and fully covered, but currently validated only as a weak geography-echo signal — not as the economic-narrative source the E_macro proposal describes. The next decision point is testing its correlation against real economic variables once Sources B/E are ingested.

2. Methodology

2.1 Ingestion Pipeline

Script: `ingest_source_a.py`

- 1. Authenticate against the Wikimedia Enterprise API.
- 2. Fetch the full article for each of ~3,222 US counties, county-equivalents, and Puerto Rico municipios (list derived from the Census Gazetteer counties file, cached in `county_crosswalk.parquet`).
- 3. Isolate the lead/introductory section only — Wikimedia Enterprise returns full HTML split into `<section data-mw-section-id="N">` blocks per MediaWiki's Parsoid model; section 0 is the lead. Infobox, later sections, and citation markers are dropped.
- 4. Embed the cleaned intro text with **BAAI/bge-m3** (1024-dim) via sentence-transformers, run on CPU (MPS/GPU auto-selection was slower for this model on short inputs).
- 5. L2-normalize each embedding (so cosine similarity reduces to a dot product) and write to `data/source_a_embeddings.parquet`.

Per-county failures (article not found, empty intro) are logged and skipped rather than aborting the run; only an authentication failure stops the whole pipeline. Coverage gaps were closed in two backfill passes: 37 Virginia independent cities (`backfill_virginia_cities.py`) and 19 remaining Census-Gazetteer-vs-Wikipedia-title mismatches (`backfill_remaining_19.py`), bringing coverage to all 3,144 US counties/county-equivalents.

2.2 Embedding Model Choice: bge-m3 vs. a Frozen LLM Encoder

The project's specs (`E_macro_extendedProposal.pdf`, `macro_pre_scoping_spec.pdf`) specify bge-m3 without arguing for it over alternatives (e.g. a frozen decoder-only LLM such as Gemma). No empirical ablation has been run in this project; the rationale below is a design justification, not a tested comparison:

- **Training objective match** — bge-m3 is trained with a contrastive/retrieval objective specifically to place semantically similar texts close together under cosine similarity. A frozen decoder-only LLM is trained for next-token prediction; pooling its hidden states is a known workaround, not what it was optimized for.
- **Benchmark performance** — purpose-built embedding models in bge-m3's class lead retrieval/STS benchmarks (e.g. MTEB) at their parameter size; frozen general-purpose LLMs underperform on semantic-similarity tasks unless separately fine-tuned into an embedding checkpoint.
- **Context length** — bge-m3 natively supports up to 8,192 tokens, matching the ingestion design's chunking/mean-pooling fallback without forcing chunking on most articles.
- **Compute footprint** — bge-m3 (568M params) runs on CPU at usable throughput for ~3,144 short texts; a multi-billion-parameter decoder LLM would cost substantially more compute for the same end product (one fixed vector per county) with no generation capability actually used.
- **Single deterministic forward pass** — producing an embedding is one forward pass with pooling, fitting the project's reproducibility requirements (fixed seeds, deterministic outputs) more naturally than adapting a generation-oriented model.

Caveat: switching to a frozen LLM encoder would not obviously fix the weak differentiation this report documents (PC1 ~3.9% variance, silhouette ≤ 0.029) — frozen decoder-LLM hidden states are known to concentrate variance in a small number of high-magnitude dimensions, which tends to hurt naive similarity comparisons. The weak signal looks more like a property of the source text (generic Wikipedia county boilerplate) than something a different

backbone would resolve on its own, but this remains an inference, not a measured result, since no such ablation has been run.

2.3 LLM Boilerplate-Cleaning Pass (Adopted 2026-07-10)

A regex/corpus-frequency approach to stripping boilerplate (targeted patterns for eponym clauses, metro-area sentences, plus a frequency filter) was judged an escalating, hard-to-maintain fix — every new boilerplate phrasing would need another regex. Instead, a single local LLM call per county rewrites `raw_intro_text`, instructed to keep only county-specific facts and drop generic templated phrasing, with an explicit no-fabrication constraint.

Method: gemma2:9b via a local Ollama server, temperature 0, one prompt per county (`reembed_source_a_llm.py`). An initial prompt version fabricated content for 1 of 8 tracked-pair counties; tightening the prompt to explicitly forbid adding any fact not literally stated in the input eliminated this, confirmed across all subsequent runs — a 128-county spot-check (8 tracked + 120 random) found 0/128 hallucination flags.

Output was re-embedded with the same bge-m3 model — the embedding backbone did not change, only the text fed into it. The cleaned parquet (`source_a_embeddings_llm.parquet`) was validated against a decision gate (minimum required drop in generic-boilerplate-pair similarity) before being adopted as the new `source_a_embeddings.parquet`; the prior regex-cleaned version is preserved at `source_a_embeddings_regex_baseline.parquet`.

Before / after example — Autauga County, Alabama (281 → 158 chars, -44%):

Raw: “County in Alabama, United States County in Alabama Autauga County is a county located in the central portion of the U.S. state of Alabama. As of the 2020 census the population was 58,805. Its county seat is Prattville. Autauga County is part of the Montgomery metropolitan area.”

Cleaned: “Autauga County is located in the central portion of Alabama. As of the 2020 census the population was 58,805. It is part of the Montgomery metropolitan area.”

Duplicated nav-label boilerplate (“County in Alabama” ×2) and disambiguation notices were consistently stripped across the corpus; reduction size shrinks as article richness grows (44% for a sparse article down to ~10% for a dense one like San Francisco County). One known over-trim: the “county seat is X” sentence was sometimes dropped even when it carried a distinctive fact (e.g. a state capital).

2.4 Analysis Methods

Mantel permutation test. Tests correlation between two distance/similarity matrices (geographic distance vs. embedding cosine similarity) while respecting non-independence between pairwise observations — a naive pairwise-correlation p-value would be invalid here since every county appears in many pairs. 499 permutations, seed 42.

PCA. Principal component analysis on the full 3,144-county embedding matrix; explained-variance ratio and manual reading of top/bottom-loading counties on PC1 used to check for an interpretable theme.

K-means clustering. Run for $k=2..12$ on a 2,849-county subset (all 50-state counties with ≥ 100 characters of cleaned text, i.e. excluding near-stub articles); k selected by silhouette score.

Cross-seed stability. K-means at the selected k re-run across 5 seeds (42, 7, 123, 2024, 99); pairwise Adjusted Rand Index (ARI) across all 10 seed-pairs measures whether the same partition recurs.

Cluster-coherence permutation test. 999 random relabelings of counties to clusters (preserving cluster sizes) establish whether the observed intra-cluster geographic tightness/looseness is more extreme than chance.

3. In-Depth Findings

3.1 Dataset & Coverage

Metric	Value
Counties in embeddings parquet	3,144 (full US coverage)
Counties used for clustering / Mantel test	2,849 (50-state, non-stub subset)
Embedding model	BAAI/bge-m3, 1024-dim, L2-normalized
Text cleaning	gemma2:9b LLM boilerplate-stripping (adopted 2026-07-10)
Metadata available	FIPS code, county name, lat/lon centroid (no economic variables yet)

3.2 Geography vs. Embedding Similarity

A Mantel test finds a weak but statistically significant negative association between geographic distance and embedding cosine similarity: nearby counties are very slightly more textually similar than far-apart ones. This result is robust across every coverage snapshot and strengthened after LLM cleaning was adopted:

	n=2,793 (original)	n=2,830 (+Virginia)	n=2,849 (regex baseline)	n=2,849 (adopted, LLM-c
Mantel r	-0.0937	-0.0898	-0.1055	-0.2362
p-value	0.0020	0.0020	0.0020	0.0020

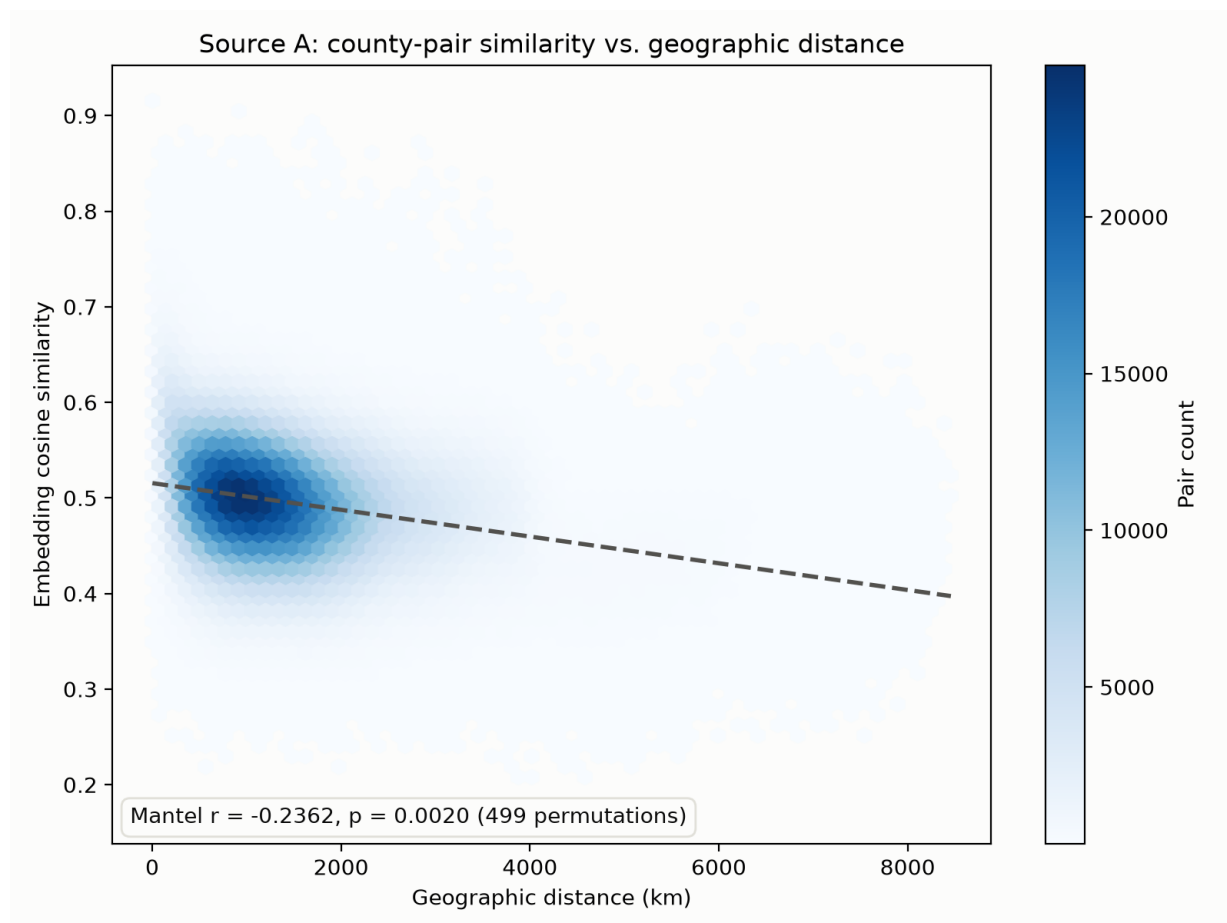


Figure 1. Hexbin of pairwise similarity vs. geographic distance, with trend line and Mantel r/p annotated. The shallow slope visually confirms a weak, not strong, relationship.

Allowed wording: “a weak, statistically significant negative association”. Avoid: any wording implying geography strongly predicts or drives embedding similarity.

3.3 Principal Component 1 — A Real but Non-Economic Theme

PC1 explains 3.9% of total variance ($n=3,144$). Unlike the superseded regex-cleaned baseline (where PC1's extremes read as generic, unthemed boilerplate), a manual read of the top/bottom 10 loading counties on the adopted LLM-cleaned embeddings finds a real, consistent pattern:

High PC1 (e.g. Elliott County KY; Swisher, Jackson, Cottle, Eastland Counties TX): cleaned text is dominated by “founded in [year]... named for [historical figure]” narrative — often a Texas Revolution soldier or 19th-century political figure — with specific dates, names, and events that survived LLM cleaning as genuinely distinctive facts.

Low PC1 (e.g. Otoe County NE; Beaver County PA; Claiborne County TN; Cassia County ID; several Iowa/North Dakota counties): cleaned text reduces to bare population + county-seat facts — near-stub articles with little distinguishing content to begin with.

Quantitative confirmation, not just an anecdotal read: **21 of the top 30 PC1 counties are Texas counties**, against Texas's 8.1% base rate in the corpus (254/3,144) — an 8.6x overrepresentation. Corpus-wide, counties whose cleaned text contains “named for”/“named after” phrasing average PC1 = 0.110 ($n=1,281$) vs. -0.076 for those that don't ($n=1,863$).

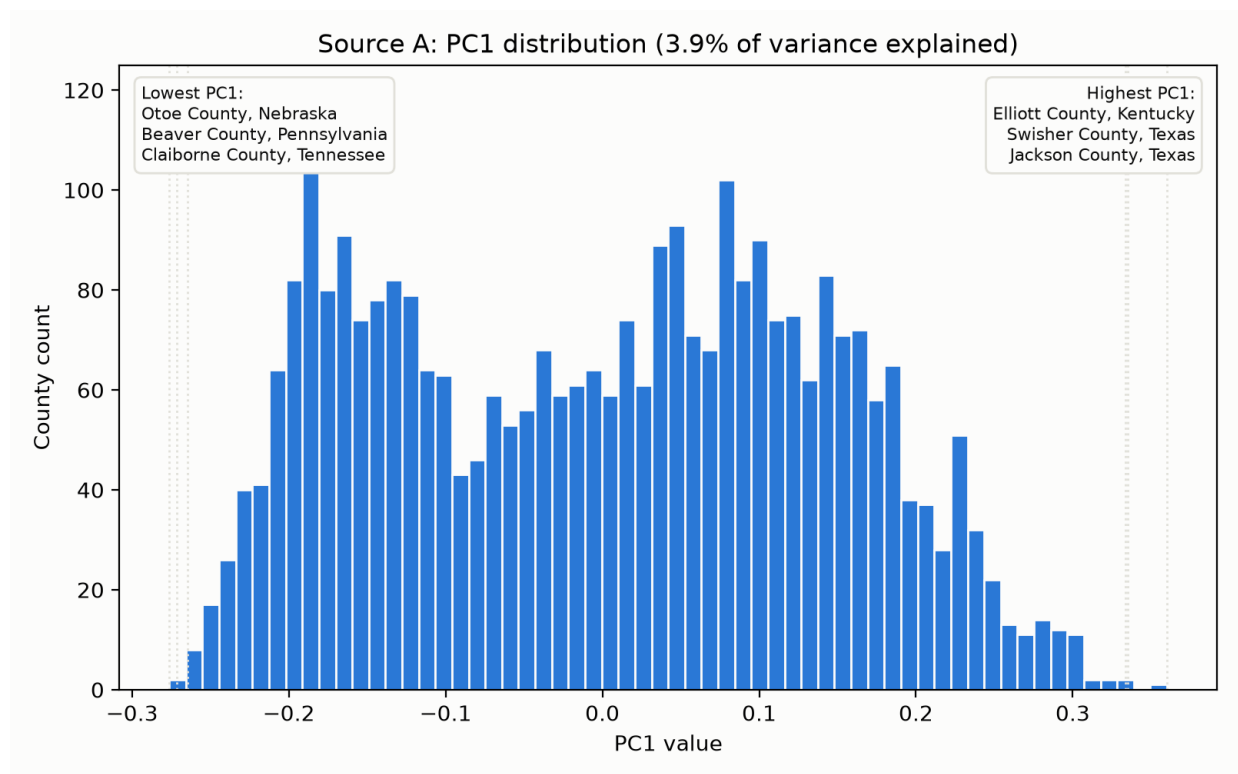


Figure 2. Histogram of PC1 values with the 3 highest/lowest counties labeled. Unimodal shape rules out a clean two-group split, but does not mean PC1 is thematically blank — see text.

Interpretation: this is a real signal, not noise — but it most plausibly reflects a **Wikipedia editorial-convention artifact** (Texas county articles disproportionately document Texas Revolution-era namesake history in enough distinctive detail that the boilerplate-stripping LLM keeps it, while many other states' articles have nothing beyond population/seat once stripped), not an economic or geographic pattern relevant to the E_macro proposal's claim. Do not read PC1 as evidence for or against the proposal's economic claim.

3.4 Clustering (K-means)

$k=2$ is selected by silhouette score across $k=2..12$, but every k 's silhouette score is low (max ≈ 0.028 , well under the ~ 0.25 threshold usually associated with real structure) — the “best” clustering is a weak fit, not a good one. On the full 2,849-county set: cluster sizes 1,420 / 1,429; cluster 0 is $\sim 12.4\%$ looser than the corpus-wide mean pairwise distance (1,635 km vs. 1,455 km), cluster 1 is $\sim 14.6\%$ tighter (1,243 km) — opposite directions, not a coherent regional split.

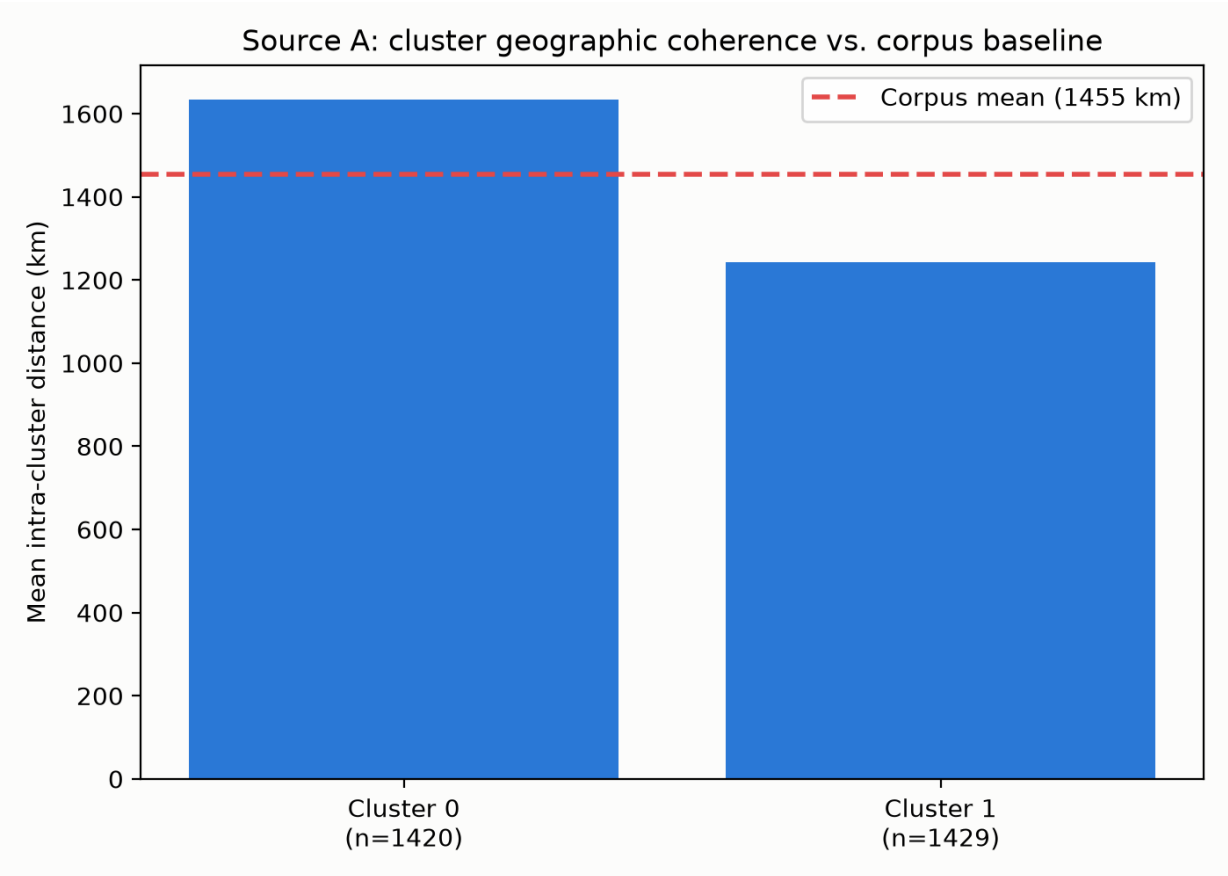


Figure 3. Mean intra-cluster distance per cluster vs. the corpus-wide baseline. Deviations run in opposite directions by similar magnitudes, blocking any “clusters correspond to US regions” claim.

Cross-seed stability: re-running K-means at k=2 across 5 seeds gives silhouette mean=0.0281, std=0.0006; pairwise ARI across all 10 seed-pairs mean=0.9813, std=0.0100, range=[0.9707, 0.9972]. The two-cluster partition is highly stable and reproducible — not an artifact of one random seed.

Cluster-coherence permutation test: 999 random relabelings (preserving cluster sizes) show both clusters' +12.4%/-14.6% deviation from the corpus mean distance is more extreme than every permutation ($p \leq 0.001$, both directions). The geographic-coherence split is statistically real, not chance — but this does not upgrade its practical size: silhouette stays ≤ 0.029 , so it is a real, reproducible, but small effect, detectable mainly because of the large sample size. Do not describe the two clusters as “regions.”

3.5 Notable Outlier Pairs

The five most similar county pairs among all top-quartile-distance pairs (≥ 75 th percentile), recomputed on the adopted LLM-cleaned embeddings:

Rank	Pair	Similarity	Distance (km)	Why similar despite distance
1	Madison Co., MT ↔ Madison Co., VA	0.876	2,866	Shared name; both reduce to the same skeleton (“founded/located in... population...
2	Jefferson Co., GA ↔ Jefferson Co., ID	0.865	2,845	Shared name and shared namesake (Thomas Jefferson) — both state “named f
3	Washington Co., ID ↔ Washington Co., WI	0.857	2,279	Shared name and shared namesake (George Washington).
4	Washington Co., KS ↔ Washington Co., NY	0.848	2,000	Shared name; same skeleton pattern as #1.
5	Lincoln Co., WA ↔ Lincoln Co., WV	0.842	3,107	Shared name (Abraham Lincoln counties recur across the corpus).

LLM cleaning shifted the mechanism behind these top pairs: previously (regex baseline) both shared-eponym and generic-boilerplate-convergence (short, minimally-elaborated articles sitting near the corpus's templated centroid) produced top-5 pairs. LLM cleaning eliminated the boilerplate-convergence mechanism from the top ranks — all 5 current top pairs are shared-name/shared-eponym only. Neither mechanism reflects real economic or geographic kinship.

3.6 Negative Result: Does Embedding More Text Help?

Tested whether embedding more of each county's article (additional body sections, or an economy-only section) differentiates counties better than the intro alone. Prototyped on a fixed 40-county sample, compared via Mantel test and pairwise similarity mean/std:

Metric	Intro-only	Lead + 3 body sections	Intro-only (same 10)	Economy-only
Mean text length	694 chars	4,532 chars (6.5x)	949 chars	697 chars
Pairwise similarity mean	0.559	0.585	0.556	0.509
Pairwise similarity std	0.068	0.054	0.069	0.072
Mantel r (geo vs. similarity)	-0.272, p=0.004	-0.372, p=0.002	-0.287, p=0.016	-0.093, p=0.494 (n.s.)

Adding body sections made counties *more* alike, not less, despite 6.5x more text — consistent with History/Geography/Demographics sections being even more templated than the lead. The economy-only variant directionally differentiated better, but only 10/40 sampled counties even had an Economy section, too small a sample to trust. A follow-up tested whether LLM-cleaning the expanded text could rescue the signal — it instead weakened it further (Mantel r dropped from -0.199 to -0.106, losing significance), because the “keep distinguishing facts” filter kept geographic anchoring for some counties and dropped it for others inconsistently. **Conclusion: closed, no further section-expansion variants planned**; if more differentiating signal is needed, the next lever is a non-Wikipedia source, not another section-selection variant.

3.7 Proposal Alignment Assessment

The `E_macro_extendedProposal.pdf` justifies Source A on the basis that Wikipedia intro text contains distinctive economic-transition language (e.g. “former rust-belt manufacturing center currently transitioning to a health-tech hub”) giving immediate semantic separation between counties, providing “narrative identity” for the Capital Flow pillar alongside Source E.

Assessment against the findings above: if economic-narrative language of the kind the proposal describes were common and distinguishing, it should surface as structure in PCA or clustering. It doesn't — PC1 explains only 3.9% of variance, and while it does carry a real theme (§3.3), that theme is founding/namesake narrative presence vs. demographic-stub content, not economic-transition narrative. K-means finds no cleanly separated structure (silhouette ≤ 0.029) beyond the weak geographic effect. The section-expansion negative results (§3.6) reinforce this: embedding more article text made counties more similar, consistent with county articles being dominated by templated incorporation/geography/demographics prose rather than distinctive economic narrative.

Not yet tested: correlation between Source A embedding distance and actual economic variables (Source E's capital-gains/*W*-2 ratio, Source B's industry location quotients). No economic ground truth is in-repo yet, so this is an open gap, not a resolved negative.

Verdict: Source A's specific mechanism in the proposal — that intro text carries salient economic-transition language usable as narrative context for Capital Flow — is not supported by the unsupervised analysis. The corpus instead reads as generic boilerplate with a very small geographic echo. This does not mean Source A should be scrapped (the geographic signal is real, and correlation with actual economic labels is untested), but it should not yet be treated as validated for the proposal's stated role.

Recommended before using Source A in `E_macro` as currently proposed:

- Test embedding distance against real economic variables (Source E's ratio, Source B's location quotients) once those sources are ingested — this is the proposal's actual claim, and it remains untested.
- If that correlation is also weak, consider targeted extraction (sentences containing transition/industry keywords) rather than whole-intro embedding, since naive full-text expansion already failed (§3.6).

3.8 Limitations

- No alternative clustering method (e.g. hierarchical, DBSCAN) has been tried as a robustness check on the K-means finding.
- No economic ground truth is in-repo yet — correlation of Source A against actual economic variables (Sources B/E) is the single most important open validation step and has not been run.
- Model-choice rationale (bge-m3 vs. a frozen LLM encoder, §2.2) is a design justification, not a tested ablation on this corpus.

3.9 Reference Numbers

Metric	Value
Counties in embeddings parquet	3,144
Counties used for clustering/Mantel test	2,849
PC1 explained variance ratio	0.0394
Selected k (silhouette)	2
Silhouette score at selected k	0.0284

Metric	Value
Mantel r	-0.2362
Mantel p-value	0.0020
Mantel permutations	499
Corpus mean pairwise distance (km)	1,455
Cluster 0 size / mean intra-cluster km	1,420 / 1,635
Cluster 1 size / mean intra-cluster km	1,429 / 1,243

Source: analysis-output/stats.json, source-a-findings.md, figures/source-a-numeric-summary.md (MacroEmbeddings repo, regenerated 2026-07-10 against the adopted LLM-cleaned embeddings).